Journal of Infrastructure Preservation and Resilience

RESEARCH Open Access

Integrating multi-source data for life-cycle risk assessment of bridge networks: a system digital twin framework

Ziheng Geng[1], Chao Zhang[2], Yishuo Jiang[3], Dora Pugliese[1] and Minghui Cheng[1,4*]

Abstract

Bridges are critical infrastructure assets that face a variety of stressors throughout their service life, requiring a life-cycle approach to assess their risk profile. Recent advancements in sensing and monitoring technologies provide a powerful data foundation to improve the accuracy of life-cycle risk assessment (LCRA). However, existing works that incorporate data for probabilistic risk assessment typically focus on individual bridges and rely on single-source data, limiting their scope and applicability. To this end, a system digital twin (SDT) framework based on Bayesian network (BN) is proposed to integrate multi-source data for LCRA of bridge networks. Specifically, the SDT can capture correlations and interdependencies across various scales, including within individual components (e.g., multiple failure modes), between components within a system (e.g., bridges along a route), and across interconnected systems (e.g., bridge and hydraulic systems). It integrates data from various sources including bridge inspections, traffic monitoring facilities, and water watch stations. A coastal bridge network in Miami-Dade County, FL, is used as an illustrative example to demonstrate how the SDT integrates multi-source data for risk assessment. Additionally, several future scenarios are hypothesized to showcase the applicability and flexibility of the proposed framework in supporting risk management for infrastructure systems.

Keywords System digital twin, Life-cycle risk assessment, Infrastructure management, System reliability, Bayesian network

Introduction

Bridges are critical assets within infrastructure systems, playing a pivotal role in ensuring the functionality and connectivity of modern communities and cities. However, these assets are vulnerable, facing a variety of stressors throughout their service life. Factors such as natural hazards, aging materials, and excessive loading can accelerate the deterioration of bridges and undermine their reliability under both normal and extreme conditions [1, 2]. This underscores the necessity for a life-cycle approach to risk assessment in bridge networks. Life-cycle risk assessment (LCRA) provides a holistic approach for assessing the evolving risks associated with a structure/system over its lifespan [3, 4]. This approach involves identifying potential hazards, estimating their probability of occurrence, and assessing their potential consequences. With the risk profile, LCRA facilitates informed decision-making in planning inspection, maintenance, repair actions to ensure long-term sustainability and resilience [5, 6]. In light of increasing impacts of climate change, adopting such a holistic approach is also essential to enhance the resilience of bridge networks, enabling them to withstand and recover from disruptions while maintaining their functionality [7].

*Correspondence:
Minghui Cheng
minghui.cheng@miami.edu
[1] Department of Civil & Architectural Engineering, University of Miami, Coral Gables, FL 33146, USA
[2] College of Civil Engineering, Hunan University, Changsha 410082, China
[3] Systems Engineering, Cornell University, Ithaca, NY 14853, USA
[4] School of Architecture, University of Miami, Coral Gables, FL 33146, USA

Springer Open

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

Recent breakthroughs in sensing and monitoring technologies have significantly improved the acquisition of multi-source data reflecting bridge conditions, thereby opening new avenues for enhancing LCRA of bridge networks. For example, non-destructive testing (NDE) techniques have been powerful tools for detecting material-level flaws in bridges. Common NDE methods, such as ground penetrating radar [8, 9], close-range photogrammetry [10], infrared thermography [8, 11], and terrestrial laser scanning [12] have demonstrated efficacy in detecting defects, rebar corrosion, and cracking. Additionally, structural health monitoring (SHM) has emerged as an advanced approach for comprehensive assessment of structural condition. SHM systems utilize diverse sensors to capture physical measurements associated with loads (e.g., bridge weigh-in-motion for traffic loads [13]), environmental stressors (e.g., acoustic sensor for scour detection [14]), and bridge responses (e.g., optical-fiber sensors for strain [15, 16] and vibration monitoring [17]). Collectively, these innovations produce an extensive and comprehensive dataset that is expected to be used by LCRA models to improve prediction accuracy and reduce model uncertainty.

With the ability to leverage diverse data sources in the real world, digital twin (DT) technology is increasingly recognized as a transformative tool for life-cycle management of infrastructure systems. Specifically, DTs are virtual representations of physical assets or systems, continuously synchronized with their physical counterparts through bidirectional information exchange [18]. The distinguishing features of DTs lie in their ability to (a) integrate observations from physical asset to dynamically update the state of digital asset and (b) leverage the updated digital state to inform decision-making for the operation, maintenance, and management of physical assets. However, the development of DTs for LCRA of bridge networks faces two critical challenges. First, bridge networks are complex systems comprising multiple interconnected sub-systems, such as bridges, transportation networks, and rivers. These sub-systems exhibit correlations and interdependencies across various scales, both within individual sub-systems (e.g., multiple structural components of a single bridge and multiple bridges along a route) and across different sub-systems (e.g., bridges subject to river scour). Therefore, a systems approach is required to effectively capture these interrelationships. Second, risk assessment of bridge networks requires the integration of heterogeneous data sources. For example, bridge inspections and river measurements inform the failure probability of bridges, while traffic volume data aids in assessing the consequences of bridge failures. These datasets typically differ in format and update frequency, underscoring the need for a data assimilation approach to seamlessly integrate multi-source data into the DT.

To address these research challenges, this paper develops a scalable computational framework for LCRA of bridge networks. The framework enables the development of a system digital twin (SDT) at a large scale (e.g., city- and county-scales), the integration of data from diverse sources and different systems, and the consideration of complex statistical correlations among the random variables and data sources within the same system and across different systems. A case study in Miami-Dade County, FL, is conducted to demonstrate the proposed framework. Specifically, data from diverse sources including bridge inspections, traffic monitoring facilities, and water watch stations are utilized to update the risk profile of the bridge network. To the best knowledge of the authors, this case study is the first DT that reaches the county scale, interlinks three systems for probabilistic modeling and risk updating, and incorporates the statistical correlations. Additionally, three scenarios are presented to illustrate the flexibility of the proposed framework, showcasing the capacity of SDT to adapt to real-world changes.

Literature review

Life-cycle risk assessment of bridge networks

Extensive efforts have been dedicated to applying LCRA to spatially distributed bridge networks within transportation systems. Risk analysis typically involves two components: (a) evaluating the failure probability of individual bridges over time, and (b) estimating the direct and indirect consequences of bridge failures. For failure probability analysis, previous studies have developed time-variant reliability analysis frameworks for various failure-inducing factors, including floods [19, 20], collisions [21, 22], traffic loads [23, 24], material deterioration [25, 26], earthquakes [27, 28], tsunamis [1, 29], storm surges [30, 31], and climate change [7, 32]. Recognizing that adjacent bridges are exposed to similar environmental stressors and traffic patterns, event-based simulations [33, 34] and random field modeling [35, 36] have been employed to model the spatial correlation of bridge failures. For failure consequence assessment, estimating indirect consequences, i.e., the cost associated with reduced functionality across the bridge network, remains a challenging task due to the need to simulate the variation in traffic flow resulting from bridge failures. Existing studies have advanced traffic assignment analyses by incorporating user decision-making behaviors [6], emphasizing extreme risk scenarios [4], and enhancing computational efficiency [37, 38].

Additionally, with the rapid advancement of machine learning, surrogate modeling techniques such as

random forests [39], deep neural networks [40, 41], and graph neural networks [42, 43] have been increasingly utilized in the reliability analysis and LCRA of bridge networks to improve computational efficiency. However, a critical limitation of the abovementioned studies lies in their reliance on simulation-based approaches, where certain model parameters are assumed to remain constant over time. Therefore, these parameters fail to capture the dynamic evolution of bridge networks and may not be the best representation of the actual conditions. It may lead to potential discrepancies between model predictions and real-world performance, limiting their applicability in engineering practice.

Digital twin for bridge asset management

Recent studies have explored the implementation of DTs in bridges asset management. For example, DTs have been developed using building information modeling (BIM) and augmented with inspection data such as 3D scanning [44], digital image correlation (DIC) [45], and weigh-in-motion (WIM) [46]. Extending these advancements, modeling enabling technologies, including geometry, measurements, simulations, and other data-driven techniques, have been integrated to develop cyber-physics DTs [47–49]. Additionally, several studies have focused on the dynamic decision-making feedback loops within DTs, employing probabilistic graphic models as the computational cores and utilizing Bayesian inference to update the digital state and inform predictive maintenance decision-making [50–52].

Despite significant advancements, existing DTs for bridge asset management suffer from two limitations. First, existing works predominately focus on developing DTs for individual bridges. However, bridges are integral components of spatially distributed transportation networks, where the failure of a single bridge can trigger cascading effects, such as traffic detours and delays. These indirect impacts frequently surpass direct consequences by several orders of magnitude [53], underscoring the necessity for modeling bridge networks as complex, interdependent systems. Second, current DTs with probabilistic modeling capabilities primarily rely on single-source data for risk updating. This reliance on siloed data source constrains the capacity of DTs to capture the dynamic evolution of real-world conditions, thereby undermining their predictive accuracy. Given the diversity and heterogeneity of real-world data, a comprehensive approach is required to synergistically integrate data from multiple sources for risk updating.

Life-cycle risk assessment of bridge networks

The assessment of bridge failure risk typically involves four steps: identifying potential failure modes, quantifying their occurrence probability, estimating the associated consequences, and determining the failure risk, as illustrated in Fig. 1.

Failure probability of bridges

Herein, two failure modes over the life cycle of a bridge are included: superstructure failure and scour failure. The probability of superstructure failure is assessed using bridge inspection data, i.e., condition ratings, following the procedures suggested by Yang and Frangopol [26]. Specifically, the superstructure condition ratings are first mapped into Markov states (MSs): ratings 9 and 8 are categorized as MS1; ratings 7, 6, 5, 4, and 3 are classified into MS2, MS3, MS4, MS5, and MS6, respectively; while ratings 2, 1, and 0 are assigned to MS7. Then, the reliability index corresponding to each MS can be calculated according to the following equations:

$$\beta(t) = \frac{\theta[MS(t)] - 1}{\sqrt{\theta[MS(t)]^2 \delta_R^2 + \delta_S^2}} \tag{1}$$

where $\beta(t)$ and $\theta[MS(t)]$ denote the reliability index (unitless) and the central safety factors (unitless) of bridge superstructure at year t. δ_R and δ_S represent the coefficients of variation (COV) for capacity and demand, determined to be 0.150 and 0.375, respectively.

$$\theta[MS(t)] = \frac{\theta_7 - \theta_1}{7 - 1}[MS(t) - 7] + \theta_7 \tag{2}$$

where θ_7 and θ_1 denote the central safety factors (unitless) for the best and worst conditions, being 1 and 4.84, respectively. $MS(t)$ denotes the Markov state of bridge

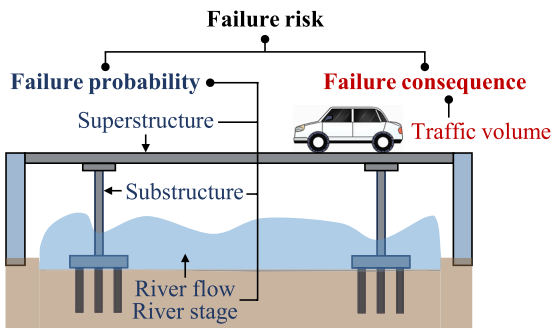


Fig. 1 Schematic diagram of life-cycle risk assessment of bridge networks

superstructure condition at year t. With the reliability index, the failure probability can be derived from the standard normal cumulative distribution function:

$$P_{super}(t) = \Phi[-\beta(t)] \tag{3}$$

where $P_{super}(t)$ denotes the failure probability (unitless) of bridge superstructure at year t and Φ represents the standard normal cumulative distribution function. As such, the superstructure failure probability can be determined for various condition ratings. Note that while condition ratings may not accurately capture the physical behavior of bridges, they are publicly accessible for all bridges across city or county scales. Future reliability analysis could be enhanced by acquiring drawings of bridges from local governments and Departments of Transportation (DOTs). With access to such data, advanced reliability assessment methodologies, such as finite element analysis [54] and analytical expressions [55] can be adopted to improve the accuracy of reliability analysis.

The probability of scour failure is estimated using bridge scour vulnerability and river depth distribution, as suggested by Stein et al. [56]. Specifically, scour vulnerability is quantified by two items in the National Bridge Inventory (NBI) database: substructure condition rating and channel protection. The depth distribution, normalized by the full depth of the river, is expressed as a dimensionless value. Its estimation involves the following steps: (a) calculate the flood drainage at a specific recurrence interval, such as 100 years, within the selected region using USGS regression equations and regard it as full flow, Q_f; (b) determine the full stage D_f by applying the stage-discharge curve provided by the USGS for the selected region; and (c) estimate the dimensionless depth ranges using available river flow or stage data via the following equation:

$$DD(t) = \frac{D(t)}{D_f} = \left[\frac{Q(t)}{Q_f}\right]^{0.6} \tag{4}$$

where $DD(t)$ denotes the depth distribution (unitless) at year t. $D(t)$ and $Q(t)$ represent river stage (ft) and river flow (ft^3/s) at year t. D_f and Q_f indicate river stage (ft) and river flow (ft^3/s) at full flow. With the scour vulnerability and depth distribution, the scour failure probability can be determined as follows:

$$P_{scour}(t) = P\{F \mid [SV(t)\ and\ DD(t)]\} \tag{5}$$

where $P_{scour}(t)$ denotes the scour failure probability (unitless) of bridges at year t. $SV(t)$ represents the scour vulnerability (unitless) of bridges at year t, which is a score from 0 to 9. F indicates the failure probability (unitless) given scour vulnerability and dimensionless depth distribution, which can be determinde by consulting the table in [56].

Failure consequence of bridges

Bridge failure can decrease link capacities within the transportation network, leading to detours and traffic delays. The economic loss associated with these detours can be divided into two primary components: the running cost and the time cost. The running cost refers to the expenses incurred from the extra distance vehicles are forced to travel. A general method to calculate running cost, accounting for various vehicle types, is as follows [57]:

$$C_{run}(t) = L_d V_{ADT}(t) T_d \left[C_{r,\,car}(1 - r_{trk}) + C_{r,trk} r_{trk}\right] \tag{6}$$

where $C_{run}(t)$ denotes the running cost (USD) of bridge failures at year t. L_d denotes the net length of the detour (km), $V_{ADT}(t)$ represents the annual average daily traffic (AADT) volume (*vpd*) at year t, T_d is the duration of the detour (days), $C_{r,car}$ and $C_{r,trk}$ are the average running cost per kilometer (USD/km) for cars and trucks, respectively, and r_{trk} denotes the ratio of average daily truck to average daily traffic (unitless). Herein, $V_{ADT}(t)$ for each bridge is approximated by data from 5 nearby short term count stations (STCSs):

$$V_{ADT}(t) = \frac{1}{5}[V_{STCS,\,1}(t)\alpha_1 + V_{STCS,\,2}(t)\alpha_2 + V_{STCS,\,3}(t)\alpha_3 + V_{STCS,\,4}(t)\alpha_4 + V_{STCS,\,5}(t)\alpha_5] \tag{7}$$

where $V_{STCS,\,1}(t)$ to $V_{STCS,\,5}(t)$ are the AADT (*vpd*) measured by 5 nearby STCSs at year t, and α_1 to α_5 are uncertainty factors (unitless) associated with the spatial interpolation of AADT [58].

The time cost includes productivity losses for both passenger vehicles and commercial trucks due to increased travel time from detours. It can be quantified by the following equation [57]:

$$C_{time}(t) = \frac{L_d V_{ADT}(t) T_d}{S_d} \left[C_{w,\,car} O_{car} (1 - r_{trk}) + C_{w,\,trk} O_{trk} r_{trk} \right] \quad (8)$$

where $C_{time}(t)$ denotes the time cost (USD) of bridge failures at year t. S_d represents the average detour speed (km/h), $C_{w,\,car}$ and $C_{w,\,trk}$ are the average wages per hour (USD/h) for car passengers and truck operators, respectively, and O_{car} and O_{trk} are the average vehicle occupancies (unitless) for cars and trucks, respectively. Detailed parameter values, particularly regarding the value of time loss for both passenger and freight vehicles, can be found in AASHTO [59]. With running cost and time cost, the bridge failure consequences can be calculated as:

$$C_f(t) = C_{run}(t) + C_{time}(t) \quad (9)$$

where $C_f(t)$ denotes the failure consequences (USD) of bridges at year t.

Failure risk of bridges

For an individual bridge, the failure risk can be quantified by the product of its failure probability and the associated consequences. Herein, two failure modes, i.e., superstructure failure and scour failure, are included and assumed to occur independently [60]. Accordingly, the failure probability of an individual bridge can be determined by:

$$P_f(t) = 1 - \left[1 - P_{super}(t)\right]\left[1 - P_{scour}(t)\right] \quad (10)$$

where $P_f(t)$ denotes the failure probability (unitless) of bridges at year t. The failure risk can be calculated as:

$$R(t) = P_f(t) C_f(t) \quad (11)$$

where $R(t)$ denotes the failure risk (USD) of bridges at year t. Under the assumption of failure independence among bridges, the total failure risk for the bridge network can be obtained by aggregating the individual failure risks of each bridge:

$$R_{network}(t) = \sum_{i=1}^{N} P_{f,i}(t) C_{f,i}(t) \quad (12)$$

where $R_{network}(t)$ denotes the failure risk (USD) of bridge network at year t, N denotes the number of bridges within the network, and $P_{f,i}(t)$ and $C_{f,i}(t)$ represent the failure probability (unitless) and failure consequence (USD) of the i-th bridge at year t, respectively.

System digital twin framework for bridge network risk assessment

Herein, a SDT framework based on BN is developed to integrate data from multiple sources for bridge network risk assessment. BNs have emerged as a pivotal enabler for implementing DT due to their capabilities in: (a) modeling complex correlations and interdependencies within infrastructure systems, (b) integrating diverse real-world data sources through Bayesian updating, and (c) performing probabilistic inference and prediction on variables of interest. The development of a BN-based SDT involves several critical steps: (a) defining the graph structure, i.e., BN structure, to capture the complex correlations within the system, (b) eliciting conditional probability tables (CPTs) to quantify dependencies between variables, and (c) specifying data nodes and performance nodes to enable automated Bayesian inference, as shown in Fig. 2. This section begins with an introduction of the concepts of BN, followed by a detailed explanation of how the graph structure is established for both bridge-level and network-level risk assessment. Lastly, it elaborates the classification of nodes and provides a streamlined approach for implementing probabilistic inference.

Concepts of Bayesian network

BN is a probabilistic graphic model that encodes the dependencies and causal relationships among variables [61]. It is structured as a directed acyclic graph (DAG), where each node denotes a random variable and each directed edge signifies a statistical dependency between two variables. A BN can be formulated as a joint probability distribution multiplied by the marginal probability distributions and conditional probability distributions. Taking an example of the BN with four nodes in Fig. 3, nodes A and B are root nodes, described by marginal probability distributions $P(A)$ and $P(B)$, respectively. Node C, as a child of nodes A and B, is described by a conditional probability distribution $P(C \mid A, B)$. Similarly, node D, as a child of node B, is described by a conditional probability distribution $P(D \mid B)$. Using these marginal and conditional probability distributions, the joint probability distribution of the BN can be described by:

$$P(A, B, C, D) = P(A)\, P(B)\, P(C \mid A, B)\, P(D \mid B) \quad (13)$$

This joint probability distribution underpins the BN, supporting probabilistic inference across complex dependencies within the model.

The construction of a BN involves defining the graph structure, which specifies the dependencies or causal

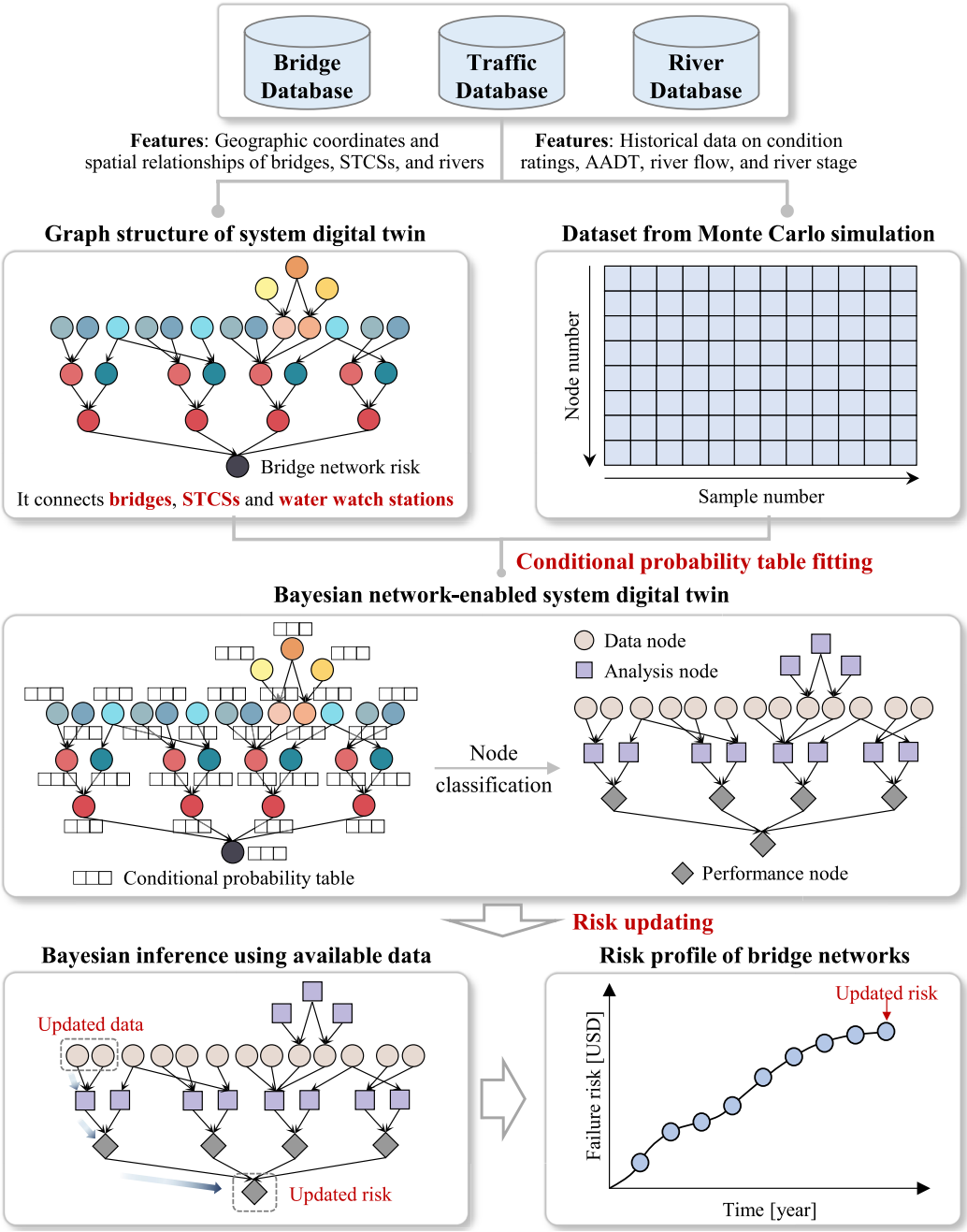


Fig. 2 Development of system digital twin for life-cycle risk assessment of bridge networks

relationships among variables, and estimating the CPTs that encapsulate the probabilistic dependencies of each node on its parent nodes. The graph structure can be crafted through domain expertise, empirical data, or a synthesis of both. Structure learning algorithms, such as constrained-based, score-based, and hybrid algorithms, can be utilized to identify an optimal structure [62–64]. Once the graph structure is determined, the next step is to assign CPTs to each node. The CPTs can be estimated from data using techniques such as Expectation–Maximization (EM) or maximum likelihood estimation, particularly in cases of incomplete or partially observed datasets [65, 66]. With the constructed BN, Bayesian inference can be conducted to derive the posterior probabilities for a set of variables of interest based on the evidence provided by another set of variables. This capacity enables the BN to

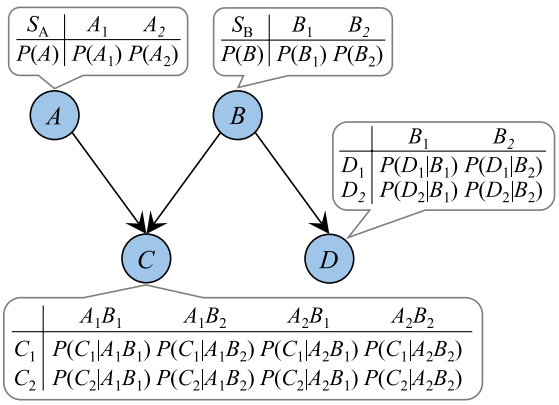


Fig. 3 Schematic diagram of a Bayesian network with four nodes

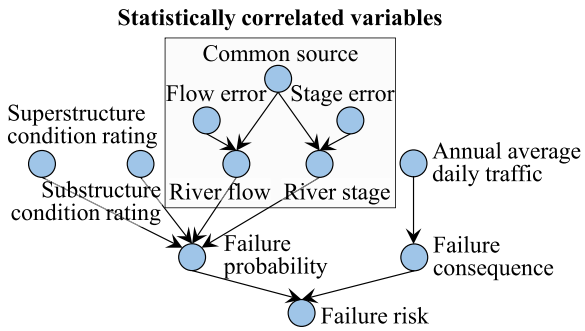


Fig. 4 Graph structure of system digital twins for bridge-level risk assessment

align closely with the core principles of DTs, where physical measurements can be incorporated into the digital space to update the state of virtual representation.

Graph structure of system digital twins for bridge-level risk assessment

The graph structure of SDTs for individual bridge risk assessment is illustrated in Fig. 4. As outlined in Sect. 3, the failure of bridges includes two modes: superstructure failure and scour failure. The probability of superstructure failure is assessed using superstructure condition rating, thus an edge is added from the superstructure condition rating node to the failure probability node. In case of scour failure, the failure probability depends on both the substructure condition and the hydraulic factors, thus three edges are added from the substructure condition rating node, river flow node, and river stage node to the failure probability node. One significant consequence of bridge failure is vehicle detour, thus an edge is added from the AADT node to the failure consequence node. The risk of bridge failure can be obtained by multiplying the failure probability with its associated consequences, thus two edges are added from the failure probability node and the failure consequence node to the failure risk node.

Additionally, a clustered graphical structure is utilized to capture statistical correlations among variables. Typically, such correlations are represented by drawing vectors from a Gaussian random field. However, exact modeling of random fields requires edge connections among each pair of correlation variables, resulting in a densely connected graph structure. This intensive connectivity significantly escalates computational burden, as the memory usage of BN increases exponentially with the number of edges [67]. Herein, an approximation method is employed to enable computationally tractable modeling of statistically correlated random variables [68]. Specifically, correlated variables are linked to a common source node, symbolizing their mutual origin in a shared random field. This configuration is exemplified in Fig. 4, where river flow node and river stage node are connected to the common source node. To further improve the modeling accuracy, error nodes are connected to each random variable, as illustrated by the flow error node and stage error node in Fig. 4. More details about the mathematical derivation and validation of modeling correlated random variables can be found in Bensi et al. [68].

Graph structure of system digital twins for network-level risk assessment

The construction of graph structure of SDTs for network-level risk assessment necessitates the incorporation of spatial geometry of various components across the network. This implies that whether the river flow, river stage, and AADT nodes can be linked to bridge failure probability and failure consequence nodes is dictated by their spatial relationship. Specifically, the linkage of river flow and stage nodes to bridge failure probability node needs to satisfy: (a) the river monitored by the water watch station flows through the bridge, and (b) the distance between the station and the bridge is less than a predefined threshold. Similarly, the linkage of AADT nodes to bridge failure consequence node requires that: (a) the bridge lies along the route monitored by the STCS, and (b) the distance between the STCS and the bridge falls below a given threshold. Additionally, beyond the statistical correlation between river flow and stage for individual river gages, the graph structure also captures the interdependencies of flow and stage among adjacent river gages. When these river gages measure the same river and their distance is below a certain threshold, the flow and stage nodes associated with these gages are modeled using a clustered graphical structure, i.e., sharing a common source node.

A hypothetical road network is utilized to demonstrate the construction of graph structure for network-level

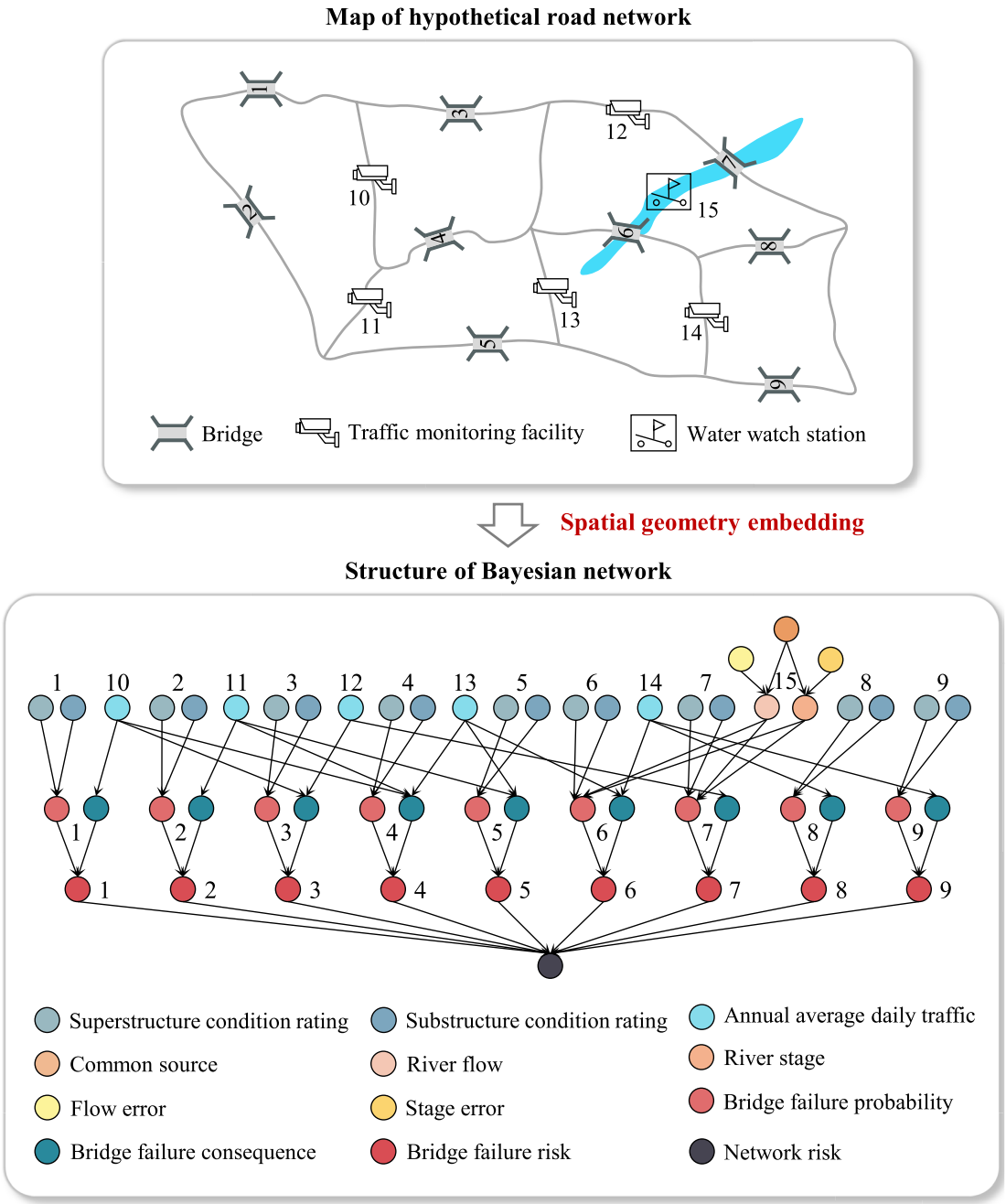


Fig. 5 Graph structure of system digital twins for network-level risk assessment

risk assessment, as depicted in Fig. 5. To better showcase how variables are connected, the figure shows the IDs for bridges, traffic monitoring facilities, and water watch stations. Specifically, the road network consists of 9 bridges, 5 traffic monitoring facilities, and 1 water watch station. First, a superstructure condition rating node and a substructure condition rating node are configured for each bridge, linking each of them to the corresponding failure probability nodes. Next, the AADT nodes adjacent to each bridge are examined for compliance with the linkage requirements, and if so, these nodes are connected to the associated failure consequence nodes. For instance, the AADT nodes corresponding to traffic monitoring facilities 10, 11, and 13 are linked to the failure consequence node of bridge 4. Lastly, river data from water watch station 15 is modeled using a clustered graphical structure

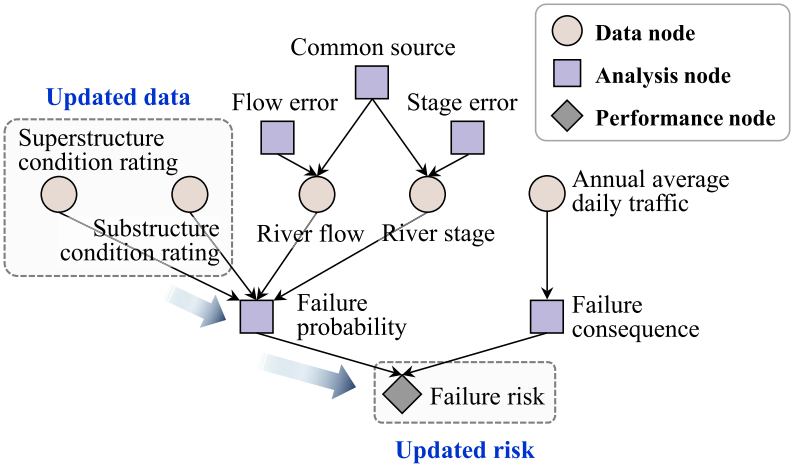


Fig. 6 Streamlined approach for implementing probabilistic inference

to capture the statistical correlation between river flow and stage. To represent the effects of river scour on bridges, the river flow and stage nodes are linked to the failure probability nodes for bridges 6 and 7. More details about the automated generation of graph structure can be found in Cheng et al. [69].

Probabilistic inference usingof system digital twins

With the constructed graph structure, a dataset needs to be created to fit the marginal or CPTs for each node of the BN. This dataset can originate from field measurements or simulations. Before being used to fit the CPTs, the dataset should be discretized, transforming continuous data into discrete states. Various discretization strategies can be applied depending on the characteristics of the dataset, for example, equal-width and equal-frequency binning allow for uniform segmentation of data, while quantile-based discretization is typically used to emphasize specific data ranges [70]. Herein, Monte Carlo (MC) simulation is utilized to generate the dataset. For discretization, equal-width and quantile-based strategies are applied selectively to handle various variables within the dataset. More details about the MC simulation and the discretization processes will be elaborated in the section of case study.

An important property of BNs is their ability to perform inference on variables of interest when data are fully or partially available. To facilitate automated inference over time, BN nodes are classified into data nodes, analysis nodes, and performance nodes based on their functions, as shown in Fig. 6. Data nodes include variables such as superstructure and substructure condition ratings, river flow, river stage, and AADT. These variables can be acquired at different frequencies from diverse monitoring devices in real-world conditions. Analysis nodes include common source, flow error, stage error, failure probability, and failure consequence, which are intermediate variables involved in the Bayesian inference process. Performance node refers to the failure risk at the individual bridge level and network risk across the bridge network, serving as key indicators of the component/network performance. At each inference, data nodes are updated based on the available data sources and Bayesian inference is performed to update the probability distribution of performance nodes. Please note that an illustrative example of a BN for monitoring the life-cycle risk of a bridge network is available in a previous study by the authors [71].

Case study: A bridge network in Miami-Dade County

Graph structure of system digital twins

The bridge network in Miami-Dade County, FL, is used as a case study to illustrate the practical application of the proposed SDT framework. The graph structure is constructed using three databases: the bridge database, traffic database, and river database. The bridge database is derived from the NBI database [72]. The features utilized to construct graph structure include the geographic coordinates of each bridge, the intersecting rivers, and the routes on which the bridges are located. Figure 7 shows a plan view of bridge locations within Miami-Dade County. The traffic database is provided by the Florida Department of Transportation (FDOT) [73], containing details on geographic coordinates of each portable traffic monitoring site and associated roadway numbers. The river database is acquired from the U.S. Geological Survey (USGS) National Water Resources Dashboard [74], providing the geographic coordinates of each river gage and monitored stream names. Note that a preprocessing

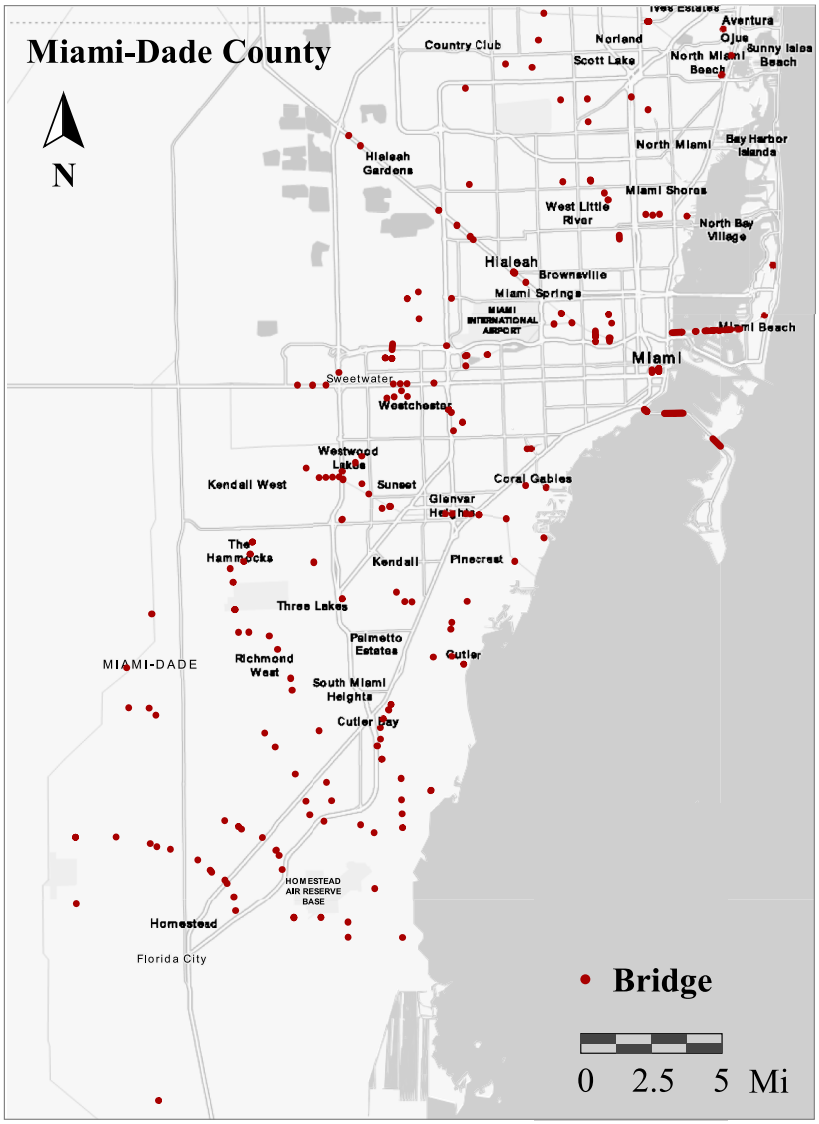


Fig. 7 Geographic location of bridges within Miami-Dade County

step is conducted to standardize coordinate formats across these databases.

Then, the AADT nodes and river nodes are evaluated to determine whether they satisfy the requirements for linkage to the bridge nodes. Specifically, the river flow and stage nodes can be linked to the bridge failure probability node if the river intersects the bridge and the distance between the river gage and the bridge is less than 5 km. For the AADT nodes, the 5 nearest STCSs monitoring the roadway where the bridge is located are selected to connect to the bridge failure consequence node. Additionally, data from two river gages are considered statistically correlated if they measure the same river and the distance between them is less than 2 km. These river flow and stage nodes are modeled using a shared common source node. Following the above steps, the graph structure for monitoring the life-cycle risk of the bridge network in Miami-Dade County is constructed, as shown in Fig. 8. The graph integrates 1,034 bridges, 1,261 STCSs, and 10 river gages within Miami-Dade County, containing a total of 6,478 nodes and 10,584 edges.

Conditional probability tables fitting

MC simulation is conducted to generate a dataset to fit the CPTs of the BN. The procedure begins with determining data ranges for certain random variables. Specifically, the superstructure and substructure condition

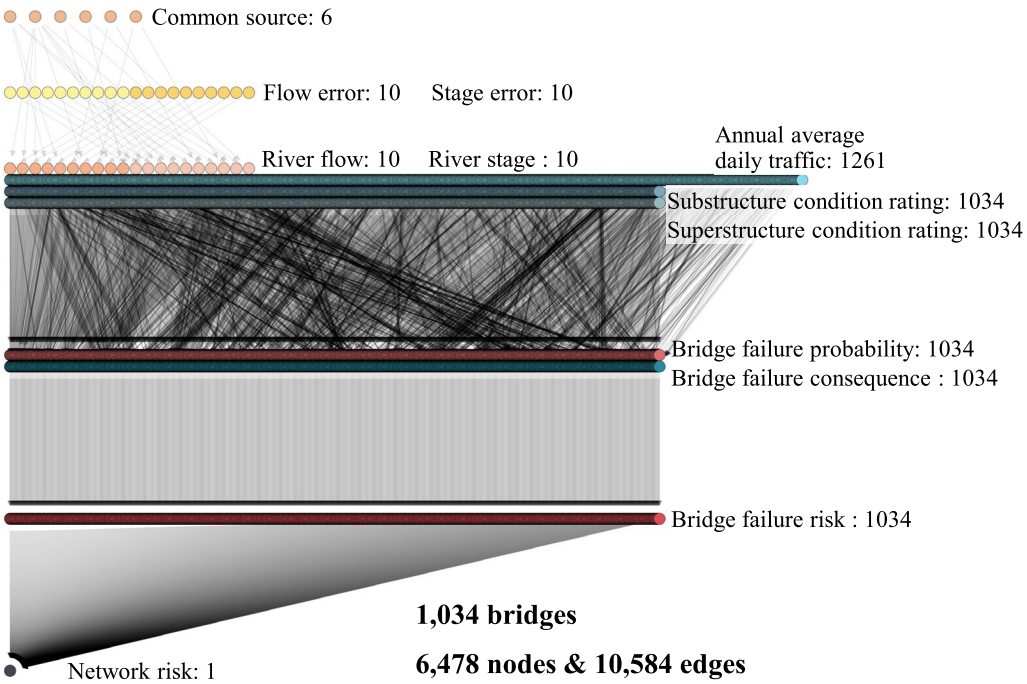


Fig. 8 Graph structure for bridge network risk assessment in Miami-Dade County

ratings are converted to MSs ranging from 1 to 7, assuming a uniform distribution. For river flow and stage, data ranges are determined according to historical river gage measurements obtained from the USGS website [74]. Figure 9 shows data from three representative river gages in Miami-Dade County. It shows that despite the presence of outliers, river flow and stage typically fluctuate within ± 30% of their mean values and are normally distributed. These findings are used to define random variables of river flow and stage in the MC simulation. Similarly, data ranges of AADT are determined using historical records from the FDOT open data hub [73]. The analysis of six representative STCSs indicates that AADT varies within a range of 0.8 to 1.8 times the initial values, as shown in Fig. 10. This variability is incorporated into the definition of random variables for AADT in the MC simulation.

To calculate the failure probability of bridges, samples are uniformly drawn from the superstructure and substructure condition ratings, as well as river flow and stage. In parallel, AADT samples are uniformly drawn from 5 nearby STCSs to calculate the failure consequence of bridges. Herein, the uncertainty factor associated with the spatial interpolation of AADT is assumed to follow a triangular distribution with a mode of 1.0, a lower bound of 0.6, and an upper bound of 1.4. Other parameters relevant to the calculation of failure consequence are taken from Cheng and Frangopol [75], summarized in Table 1. The failure risk is calculated as the product of failure probability and failure consequence. Please note that the data for common source node, flow error node, and stage error node are derived using Cholesky factorization of river flow and stage data [68].

The dataset must be discretized before being utilized to learn the CPTs of the BN. Specifically, the superstructure and substructure conditions of bridges are represented by MSs, which are inherently discrete variables. For continuous variables such as common source, river flow, river stage, flow error, stage error, and AADT, an equal-width discretization is employed, dividing the range of each variable into 6 uniform intervals. In contrast, variables relevant to the calculation of risk, including bridge failure probability, bridge failure consequence, bridge failure risk, and network risk, are discretized using a quantile-based strategy. This strategy leverages 25th, 50th, 75th, 85th, and 95th percentiles to capture critical variation in data distribution and enable a more granular representation of high-risk scenarios. With the discretized dataset, parameter learning is conducted to fit the CPTs of the BN.

Probabilistic inference for network risk

A future scenario is hypothesized to illustrate how the SDT framework performs risk inference using data from various sources. Herein, the data sources include bridge inspection reports, traffic monitoring facilities, and water

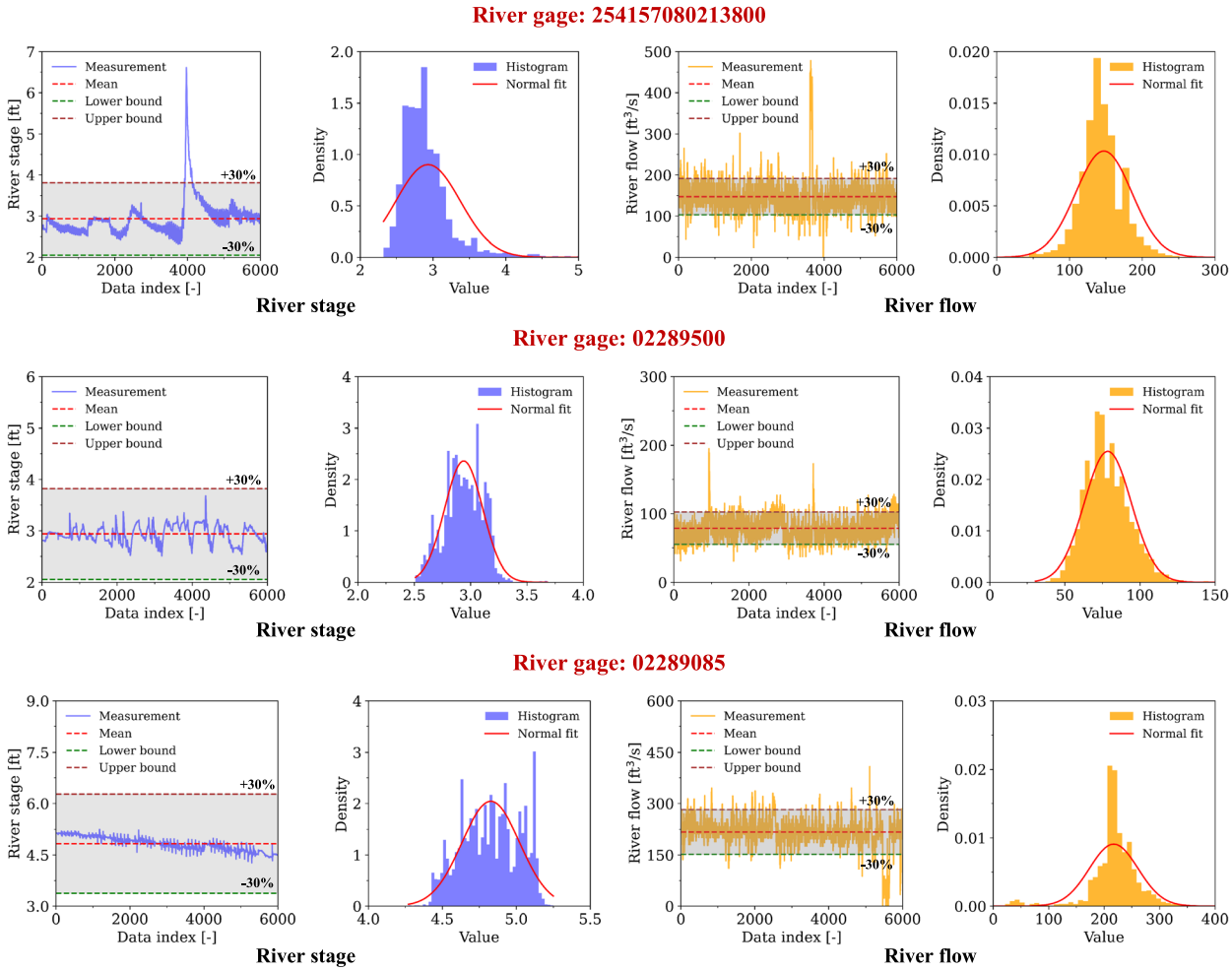


Fig. 9 Historical river gage measurements obtained from the USGS website [74]

watch stations. The initial data are derived from realistic measurements: the NBI database for superstructure and substructure condition ratings, the FDOT for AADT data corresponding to each STCS, and the USGS for river flow and stage at gage stations. To simulate the deterioration of bridge structural conditions, the superstructure condition and substructure condition of bridges are modeled using Markov chain with the following transition matrices [26], respectively:

$$\mathbf{T}_{super=}\begin{bmatrix} 0.9719 & 0.0233 & 0.0036 & 0.0012 & 0 & 0 & 0 \\ 0 & 0.9844 & 0.0113 & 0.0036 & 0.0004 & 0.0003 & 0 \\ 0 & 0 & 0.9823 & 0.0152 & 0.0020 & 0.0005 & 0 \\ 0 & 0 & 0 & 0.9910 & 0.0070 & 0.0020 & 0 \\ 0 & 0 & 0 & 0 & 0.9922 & 0.0078 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.9950 & 0.0050 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{14}$$

$$\mathbf{T}_{sub=}\begin{bmatrix} 0.9164 & 0.0801 & 0.0035 & 0. & 0 & 0 & 0 \\ 0 & 0.9885 & 0.0107 & 0.0006 & 0.0003 & 0. & 0 \\ 0 & 0 & 0.9842 & 0.0130 & 0.0021 & 0.0007 & 0 \\ 0 & 0 & 0 & 0.9874 & 0.0106 & 0.0020 & 0 \\ 0 & 0 & 0 & 0 & 0.9915 & 0.0085 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.9848 & 0.0152 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{15}$$

Additionally, the river flow, river stage, and AADT data are assumed to increase with a growth coefficient following a normal distribution with a mean of 1% and a variance of 1%. Several representative sets of simulated data are depicted in Fig. 11.

The simulated data are incorporated into the SDT as evidence to perform Bayesian inference. According to the National Bridge Inspection Standards (NBIS), safety inspections for highway bridges are mandated to be conducted at least once every 24 months [76]. For bridges with components in poor condition, the inspection interval is further reduced to a maximum of 12 months.

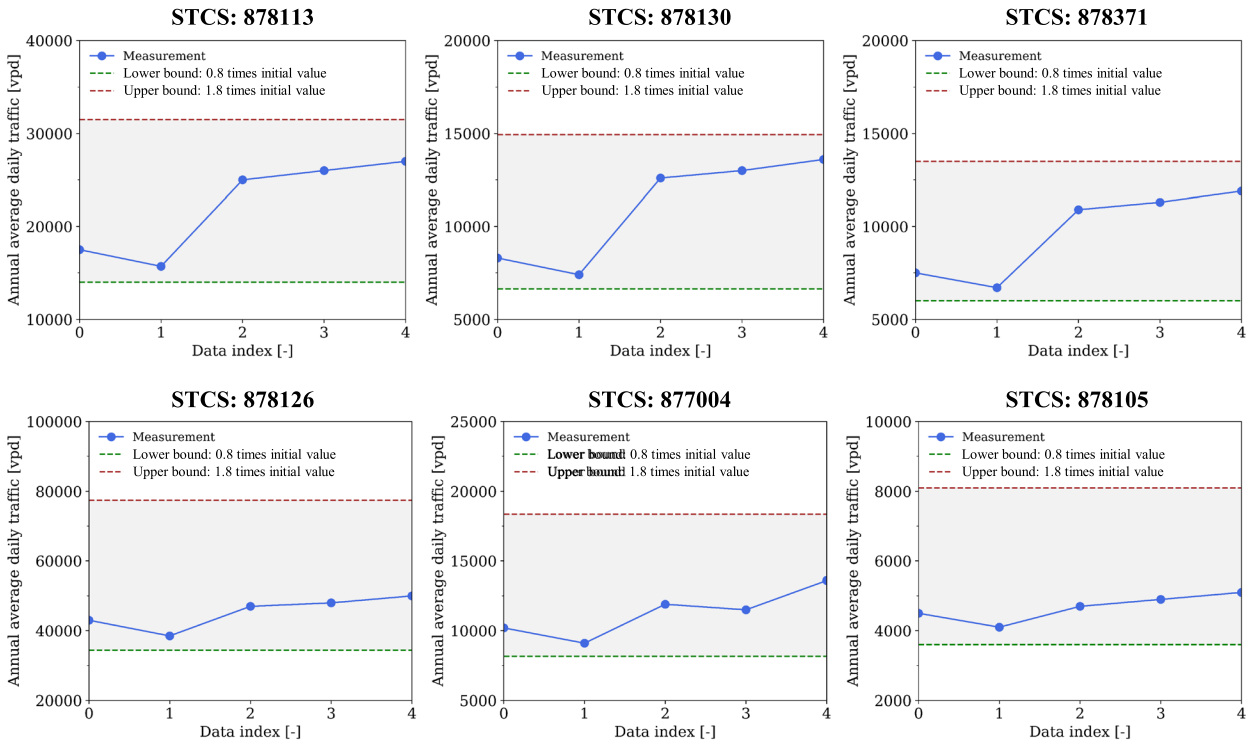


Fig. 10 Historical AADT data obtained from the FDOT open data hub [73]

Therefore, the update frequency of bridge network risk profile within the SDT is determined to be an annual basis, ensuring the timely identification and management of potential risk. To replicate real-world conditions, bridge inspections, river measurements, and traffic monitoring data are updated at various frequencies. Specifically, the bridge condition ratings are updated every two years, the AADT data is updated every three years, and river data is updated annually.

To streamline this process, an indicator matrix is developed to represent data availability each year. This matrix uses a binary index, where a value of 1 denotes data availability and 0 signifies data missing. The indicator matrix can facilitate the automated Bayesian inference for each year. The inferred network risk profile is shown in Fig. 12. Despite minor fluctuations, the network risk exhibits a general upward trend over time. This trend can be attributed to the deterioration of bridge conditions, increased river flow and stage, and rising AADT. While exhibiting an upward trend, Fig. 12 also shows that the network risk can sometimes drop, such as at years 10, 21, 24, and 27. This can be attributed to the fact that data is available at these years and Bayesian inference is performed. With the presence of data and Bayesian updating, the uncertainty associated with certain variables of interest is reduced, resulting in a smaller risk. These results underscore the ability of the proposed framework in updating the risk of bridge network using available multi-source data.

Flexibility of the proposed system digital twin framework

One of the strengths of the proposed SDT framework lies in its flexibility in accommodating the dynamic evolution of urban environments through the graphical structure of BN. As urban areas expand and transform, infrastructure systems undergo significant changes, including the construction of new bridges, the closure of aging structures, and heightened exposure to climate-related challenges, such as the increasing frequency and intensity of extreme weather events. These evolving conditions require the risk assessment framework to be flexible to adapt to such changes. The proposed framework addresses this need through its modular structure, where the CPTs are organized in an isolated and localized manner. This structure allows specific components within the BN to be independently modified or updated without reconfiguring the entire network. Consequently, the proposed framework has the potential to rapidly adapt to the complex changes in the real world, thus enhancing its utility for decision-makers. Herein, three scenarios are selected to demonstrate the flexibility of the proposed framework, as illustrated in Fig. 13.

In scenario 1, the SDT is updated to accommodate the construction of a new bridge. This process begins

Table 1 Parameters relevant to calculation of failure consequence

$c_{r,car}$ [USD/km]	$c_{r,trk}$ [USD/km]	T_d [days]	$c_{w,car}$ [USD/h]	$c_{w,trk}$ [USD/h]	O_{car} [USD/h]	O_{trk} [USD/h]	S_d [USD/h]
0.11	0.30	180	22.82	26.97	1.5	1.05	50

Note: L_d and r_{trk} are obtained from the NBI database

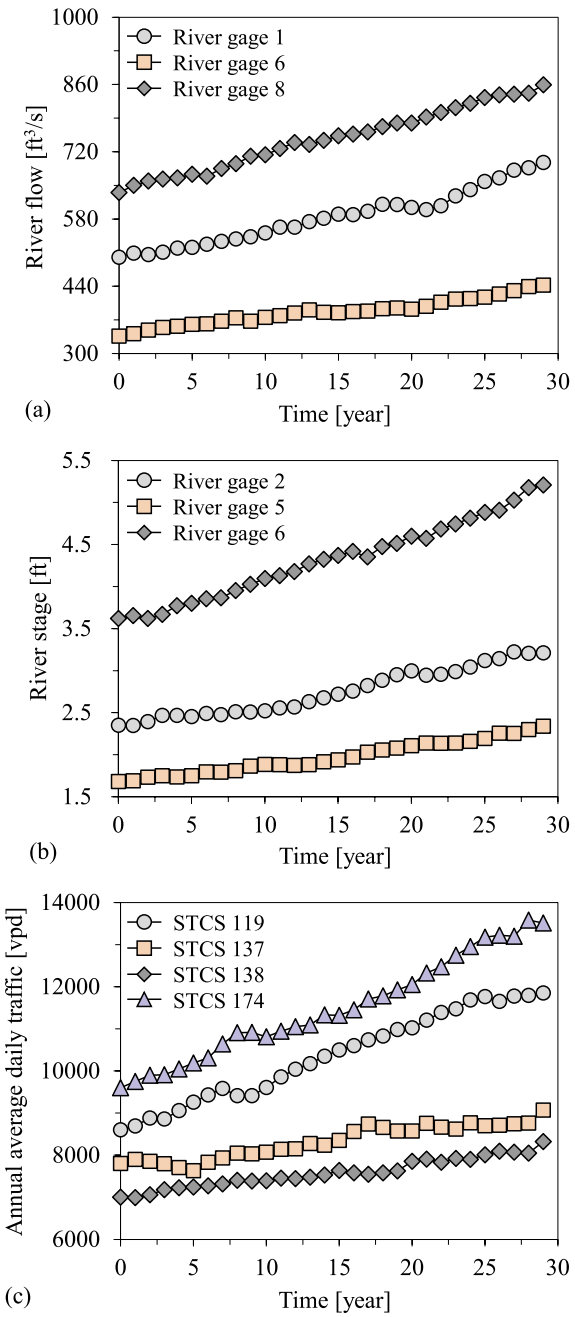


Fig. 11 Simulated data for bridge network risk inference: **a** river flow, **b** river stage, and **c** AADT

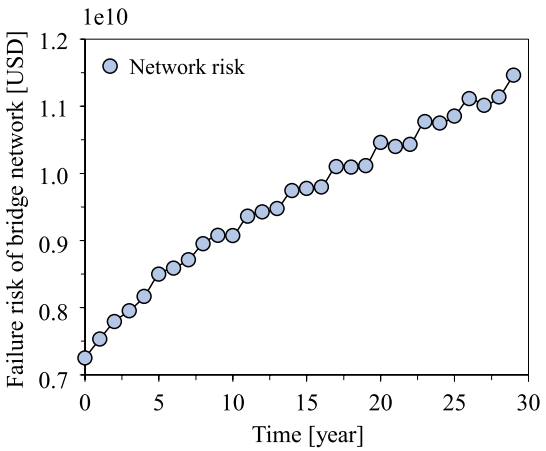


Fig. 12 Network risk profile of bridge network in Miami-Dade County

with the construction of the graph structure for the new bridge, as illustrated in Fig. 4. A critical step in this construction involves evaluating the potential connections between the new bridge nodes and existing river nodes. In this case, no such connections are established. Another essential step entails identifying 5 nearby STCSs and linking their associated AADT nodes to the failure consequence node. Following the above steps, a small-scale graph structure is constructed for the new bridge, comprising 10 nodes and 9 edges. The next step is to fit the CPTs using dataset generated through MC simulation. Due to the small scale of this BN, the computational resources required for CPT fitting are minimal. The fitted BN is integrated into the existing BN by adding nodes, edges, and CPTs, thereby producing an updated BN that incorporates the new bridge. The updated BN is employed to perform inference using simulated data in Fig. 11. The inference results, depicted in Fig. 14, reveal that the network risk increases over time, consistent with the trend observed in Fig. 12. Additionally, the failure risk of the new bridge increases over time, driven by the deterioration of its structural condition and the growth in AADT. These results demonstrate the adaptability of the proposed framework in integrating new infrastructure components within existing BNs.

Scenario 2 involves the decommissioning of an aging bridge, requiring the removal of its associated nodes from the existing BN. Specifically, the nodes representing

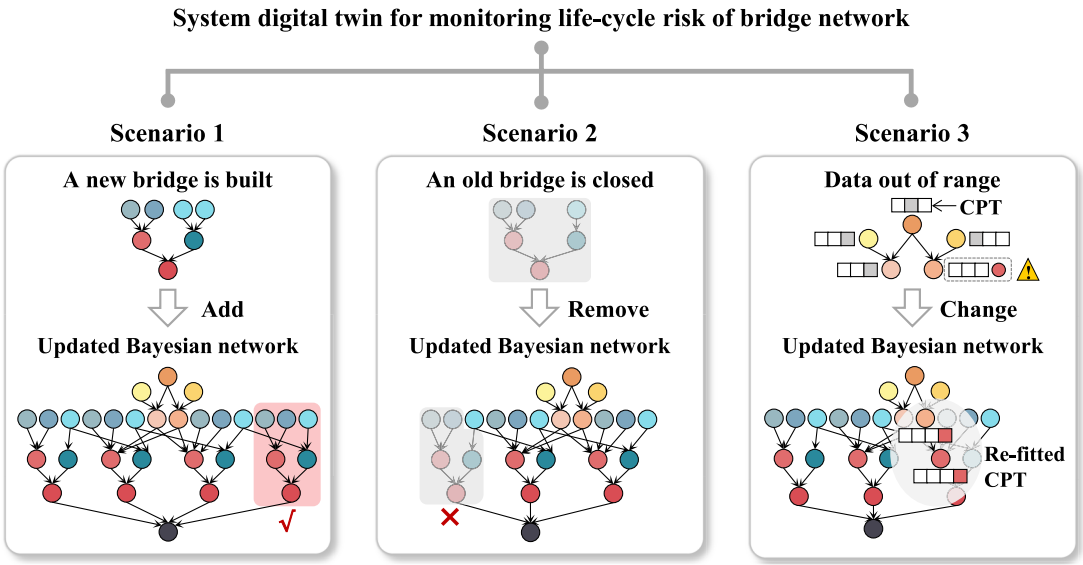


Fig. 13 Three scenarios for demonstrating the flexibility of the proposed framework

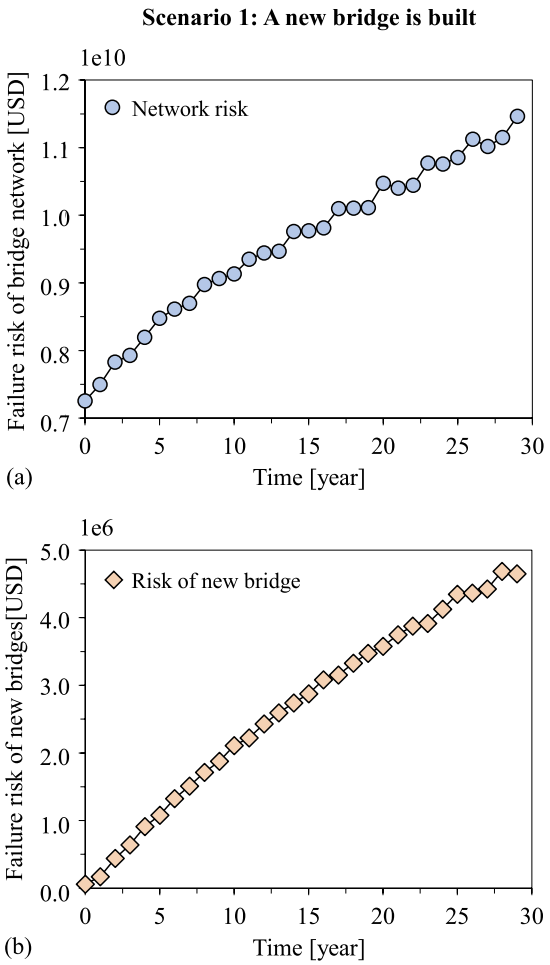


Fig. 14 Risk profile in Scenario 1: **a** bridge network risk; **b** new bridge risk

the superstructure condition rating, substructure condition rating, failure probability, failure consequence, and failure risk of the old bridge can be directly eliminated from the BN, as they are exclusive to the decommissioned bridge. However, a more nuanced treatment is required for nodes representing bridge-connected elements, such as river nodes and AADT nodes, which may be connected to other bridges within the network. If these nodes are shared by other bridges, they cannot be removed. Note that the edges associated with the removed nodes will be automatically deleted in the BN without requiring manual adjustments. After removing the nodes associated with the decommissioned bridge, the BN is updated, as illustrated in Fig. 13. Using the updated BN, Bayesian inference is performed with the simulated data in Fig. 11, yielding the results depicted in Fig. 15. The results indicate an upward trend in network risk, punctuated by localized fluctuations. This trend aligns with the patterns observed in Figs. 12 and 14(a). Consequently, the scenario highlights the capacity of the proposed framework to maintain the integrity and functionality of risk assessment even after the removal of certain decommissioned infrastructure components.

Scenario 3 is used to demonstrate the flexibility of the proposed framework in addressing challenges posed by climate change. Specifically, it considers a situation where floods or hurricanes cause river stages to rise beyond the data range of river nodes in the established BN. The proposed framework can resolve this issue by modifying the localized CPTs related to the affected river nodes, as illustrated in Fig. 13. The process begins with identifying all nodes associated with the overflowing river, including

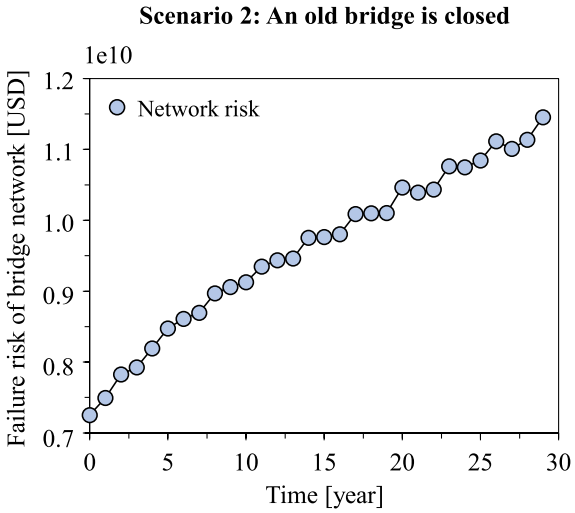


Fig. 15 Network risk profile in Scenario 2

those contributing to the calculation of failure probability for the bridges connected by the river. The sub-network is then extracted and MC simulation is conducted to generate a new dataset for recalibrating the CPTs. The dataset generation follows the previously described procedures, but with an expanded fluctuation range for river stage, adjusted from ± 30% to ± 60%. Next, the recalibrated CPTs are integrated into the existing BN, producing an updated BN. To simulate the rising river stage, a growth coefficient modeled as a normal distribution with a mean of 3% and a variance of 1% is applied, as illustrated in stage 1 from Fig. 16(b). Bayesian inference is performed using the updated BN to assess the network risk, with results shown in Fig. 16(a). Additionally, the failure risk of the bridge connected to the overflowing river is presented in stage 1 from Fig. 16(c), revealing a significant increase in risk as river stage rises.

Scenario 3 is also used to model the effect of water level fluctuations induced by climate change. Specifically, the fluctuations in river stage can be divided into four representative stages: rapid rise, slight fluctuation, sharp fall, and significant fluctuation [77], as shown in Fig. 16(b). These fluctuating river stages are used as input to update the failure risk of bridges over a 30-year period, as illustrated in Fig. 16(c). The results reveal that the failure risk increases dramatically during sudden surges in river stage. When the river stage fluctuates slightly at high levels, the failure risk continues to rise due to the progressive deterioration of superstructure and substructure conditions, but the growth rate slows down. In contrast, during periods of decreasing river stage, the failure risk remains relatively stable. This result arises from the counterbalance between increased risk due to structural

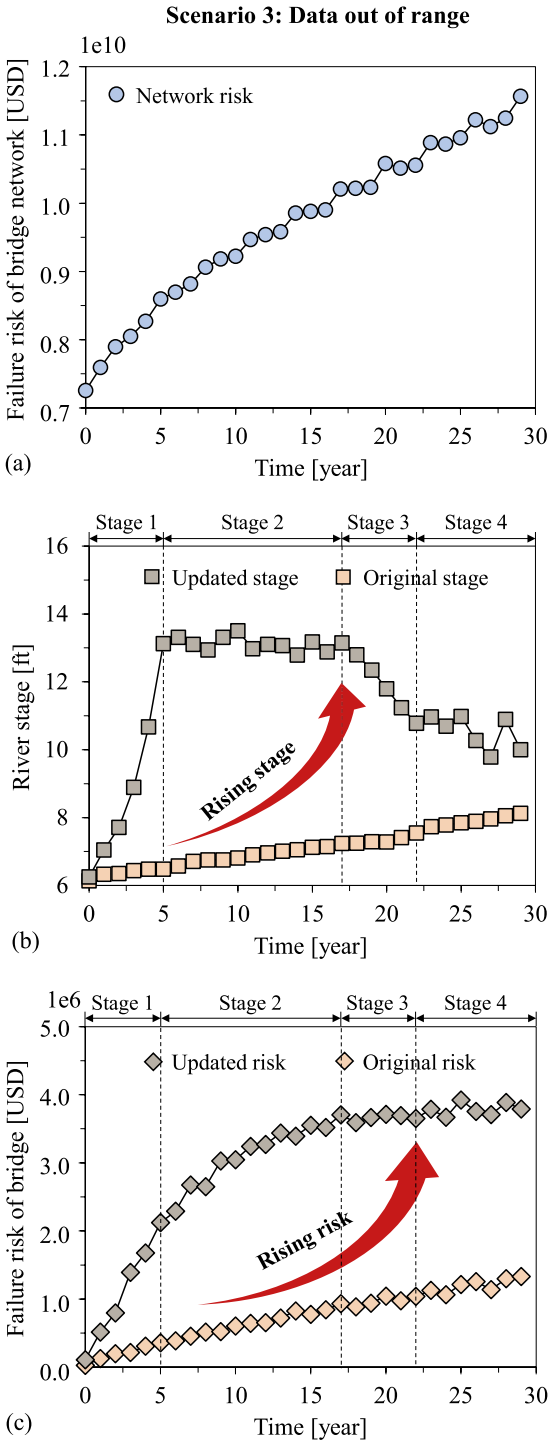


Fig. 16 Results in Scenario 3: **a** bridge network risk, **b** comparison between the rising river stage and original river stage, **c** comparison between the rising bridge risk and original bridge risk

deterioration and reduced risk from river scour. In cases where river stage fluctuates significantly, the failure risk varies correspondingly. These findings demonstrate the

effectiveness of the proposed framework in capturing variation in bridge failure risk associated with climate change-induced water level fluctuations. The framework can provide valuable insights for decision-makers, supporting the development of predictive maintenance strategies to mitigate potential failures.

Limitations and future work

Although the proposed framework demonstrates feasibility and applicability in assessing life-cycle risk for bridge networks, it suffers from certain limitations. First, a significant challenge is the storage and computational efficiency associated with large-scale BNs. For example, the BN in this study comprises 6,478 nodes and 10,584 edges, occupying 97,484.8 MB memory to store its CPTs. This presents a substantial computational burden. As the size and complexity of the network grow, the exponential increase in memory requirements can severely hinder the practical implementations of the proposed framework. (Cheng M, Geng Z, Gao HO (2024) Performance-oriented system digital twin: A Bayesian network-based framework to customize sub-digital twins for faster Bayesian inference. Struct Infrastruct E (Under review)) proposed a custom BN approach to address this challenge. It can extract a trunk network from the original large-scale BN based on available data sources and given performance measures. By reducing the size of the network, it can significantly lower the computational overhead, facilitating efficient Bayesian inference. Building on these advancements, future work should focus on seamlessly integrating custom BN with the proposed framework to develop scalable solutions that can handle the computational demands of increasingly large and complex networks.

Second, due to the limitations in available resources, the current framework has not yet been validated. Future work should focus on assessing the performance and quality of the SDT to ensure its applicability in practical engineering scenarios. Specifically, the validation can be implemented with the aid of field and laboratory testing. The proposed framework can identify a list of vulnerable bridges within the bridge network. For these identified bridges, field coring can be conducted to extract core samples from their structural components. These samples can then be tested in the laboratory to analyze their properties such as microstructural characteristics, apparent electrical resistivity, compressive strength, chloride ingress, and carbonation depths. These laboratory analyses provide quantitative metrics that reflect the physical condition of bridges. To validate the framework, the same procedures should be applied to bridges classified as low risk by the SDT. By comparing the laboratory results from vulnerable and low-risk bridges, valuable insights into the predictive accuracy and robustness of the framework can be derived.

Third, the proposed framework primarily focuses on risk assessment of bridge networks. It integrates the probability of bridge failure with the associated failure consequences to identify the risk within the bridge network. However, it does not include the system's response to address these threats. Future work should extend the framework to incorporate restoration and resilience. Resilience is a performance indicator that evaluates the system's capacity to withstand, adapt to, and recover from disruptions [54, 78, 79]. It highlights the ability of a system to recover to its previous functionality after a disruption. Restoration models, which define the recovery pattern of the system, are essential tools for quantifying resilience [80]. Incorporating these elements will enhance the utility of the framework in infrastructure management, better assisting in decision-making regarding inspections and maintenance of infrastructure.

Summary and conclusions

This paper develops a system digital twin (SDT) framework based on Bayesian network (BN) to monitor the life-cycle risk of bridge networks. A prototype graph structure of SDTs is first developed at the bridge level to capture the correlations within individual components (e.g., multiple failure modes of bridges) and to integrate data from various sources. The prototype is then extended to the network level by embedding spatial geometry of cities or counties. The resulting network-level graph structure can capture interdependencies both within a single system (e, g., bridges along a route and river gages monitoring the same river) and across interconnected systems (e.g., bridge, transportation, and hydraulic systems). With the graph structure, Monte Carlo simulation is conducted to generate a dataset to fit the conditional probability tables (CPTs) of the BN. To streamline the Bayesian inference over time, the BN nodes are classified into three types: data nodes, analysis nodes, and performance nodes. This classification facilitates automated Bayesian inference using multi-source data with different update frequencies. A real-world bridge network in Miami-Dade County, FL, is used as a case study to illustrate the feasibility and applicability of the proposed framework for life-cycle risk assessment. The following conclusions are drawn:

The graph structure is constructed using three databases: the bridge database, traffic database, and river database. By embedding the spatial geometry of the county, a graph structure with 6,478 nodes and 10,584 edges is created, integrating 1,034

bridges, 1,261 STCSs, and 10 river gages within Miami-Dade County. This demonstrates the scalability of the proposed SDT framework.

The graph structure utilizes a clustered graphical structure to capture the statistically correlated variables, including river flow and stage data from both individual and adjacent river gages. These correlated river flow and stage nodes are connected to a common source node, demonstrating the capacity of the SDT framework to model the complex correlations at the county scale.

A future scenario is hypothesized to showcase how the SDT integrates diverse data sources for risk inference. The bridge conditions are assumed to deteriorate over time, whereas traffic volumes and river stage/flow are assumed to increase gradually. The results reveal a general upward trend in network risk, punctuated by localized fluctuations. This demonstrates the capability of the proposed framework in updating the risk of bridge network using available multiple-source data.

Three representative scenarios are selected: the construction of new bridges, the closure of aging bridges, and data out of range. The proposed framework effectively addresses these changes by modifying localized CPTs without reconfiguring the entire network. The updated BN retains its integrity and functionality for risk assessment across all scenarios. The results show the flexibility of the proposed framework in accommodating the dynamic evolution of urban bridge networks.

Acknowledgements
The authors are grateful for the financial support received from US Department of Transportation Tier 1 University Transportation Center CREATE Award No. 69A3552348330.

Authors' contributions
Z.G.: Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. C.Z.: Writing – review and; editing. Y.J.: Software, Data curation. D.P.: Software, Data curation. M.C.: Writing – review & editing, Methodology, Conceptualization.

Data availability
Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Competing interests
The authors declare no competing interests.

Received: 14 December 2024 Revised: 29 January 2025 Accepted: 5 February 2025
Published online: 27 February 2025

References

1. Akiyama M, Frangopol DM, Ishibashi H (2020) Toward life-cycle reliability-, risk- and resilience-based design and assessment of bridges and bridge networks under independent and interacting hazards: emphasis on earthquake, tsunami and corrosion. Struct Infrastruct E 16:26–50
2. Frangopol DM, Dong Y, Sabatino S (2017) Bridge life-cycle performance and cost: analysis, prediction, optimization and decision-making. Struct Infrastruct E 13:1239-1257
3. Frangopol DM, Kong JS, Gharaibeh ES (2001) Reliability-based life-cycle management of highway bridges. J Comput Civ Eng 15:27–34. https://doi.org/10.1061/(ASCE)0887-3801(2001)15:1(27)
4. Yang DY, Frangopol DM (2020) Life-cycle management of deteriorating bridge networks with network-level risk bounds and system reliability analysis. Struct Saf 83:101911. https://doi.org/10.1016/j.strusafe.2019.101911
5. Ellingwood BR, Lee JY (2016) Life cycle performance goals for civil infrastructure: intergenerational risk-informed decisions. Struct Infrastruct E 12:822–829
6. Han X, Frangopol DM (2022) Risk-based optimal life-cycle maintenance strategy for bridge networks considering stochastic user equilibrium. ASCE-ASME J Risk Uncertainty Eng Syst, Part A: Civ Eng 8:04022011. https://doi.org/10.1061/AJRUA6.0001222
7. Liu L, Yang DY, Frangopol DM (2020) Network-level risk-based framework for optimal bridge adaptation management considering scour and climate change. J Infrastruct Syst 26:04019037. https://doi.org/10.1061/(ASCE)IS.1943-555X.0000516
8. Abu Dabous S, Yaghi S, Alkass S, Moselhi O (2017) Concrete bridge deck condition assessment using IR Thermography and Ground Penetrating Radar technologies. Automat Constr 81:340–354. https://doi.org/10.1016/j.autcon.2017.04.006
9. Wang Z, Zhou M, Slabaugh GG, Zhai J, Fang T (2011) Automatic detection of bridge deck condition from ground penetrating radar images. IEEE Trans Automat Sci Eng 8:633–640. https://doi.org/10.1109/TASE.2010.2092428
10. Jiang R, Jáuregui DV, White KR (2008) Close-range photogrammetry applications in bridge measurement: Literature review. Measurement 41:823–834. https://doi.org/10.1016/j.measurement.2007.12.005
11. Omar T, Nehdi ML (2017) Remote sensing of concrete bridge decks using unmanned aerial vehicle infrared thermography. Automat Constr 83:360–371. https://doi.org/10.1016/j.autcon.2017.06.024
12. Riveiro B, González-Jorge H, Varela M, Jauregui DV (2013) Validation of terrestrial laser scanning and photogrammetry techniques for the measurement of vertical underclearance and beam geometry in structural inspection of bridges. Measurement 46:784–794. https://doi.org/10.1016/j.measurement.2012.09.018
13. Yu Y, Cai C, Deng L (2016) State-of-the-art review on bridge weigh-in-motion technology. Adv Struct Eng 19:1514–1530. https://doi.org/10.1177/1369433216655922
14. Prendergast LJ, Gavin K (2014) A review of bridge scour monitoring techniques. J Rock Mech Geotech Eng 6:138–149. https://doi.org/10.1016/j.jrmge.2014.01.007
15. Ni Y, Chen R (2021) Strain monitoring based bridge reliability assessment using parametric Bayesian mixture model. Eng Struct 226:111406. https://doi.org/10.1016/j.engstruct.2020.111406
16. Ni Y, Ye X, Ko JM (2012) Modeling of stress spectrum using long-term monitoring data and finite mixture distributions. J Eng Mech 138:175–183. https://doi.org/10.1061/(ASCE)EM.1943-7889.0000313
17. Kamariotis A, Chatzi E, Straub D (2022) Value of information from vibration-based structural health monitoring extracted via Bayesian model updating. Mech Syst Signal Pr 166:108465. https://doi.org/10.1016/j.ymssp.2021.108465
18. VanDerHorn E, Mahadevan S (2021) Digital twin: Generalization, characterization and implementation. Decis Support Syst 145:113524. https://doi.org/10.1016/j.dss.2021.113524
19. Zhang N, Alipour A (2022) Flood risk assessment and application of risk curves for design of mitigation strategies. Int J Crit Infr Prot 36:100490. https://doi.org/10.1016/j.ijcip.2021.100490
20. Loli M, Kefalas G, Dafis S, Mitoulis SA, Schmidt F (2022) Bridge-specific flood risk assessment of transport networks using GIS and remotely sensed data. Sci Total Environ 850:157976. https://doi.org/10.1016/j.scitotenv.2022.157976

21. Chen L, Qian J, Tu B, Frangopol DM, Dong Y (2021) Performance-based risk assessment of reinforced concrete bridge piers subjected to vehicle collision. Eng Struct 229:111640. https://doi.org/10.1016/j.engstruct.2020.111640
22. Liu Y, Guo X (2023) Study on risk of ship collision in bridge life-cycle based on synergetic theory. Ocean Eng 289:116148. https://doi.org/10.1016/j.oceaneng.2023.116148
23. Zhu B, Frangopol DM (2013) Risk-based approach for optimum maintenance of bridges under traffic and earthquake loads. J Struct Eng 139:422–434. https://doi.org/10.1061/(ASCE)ST.1943-541X.0000671
24. Fiorillo G, Ghosn M (2019) Risk-based importance factors for bridge networks under highway traffic loads. Struct Infrastruct E 15:113–126. https://doi.org/10.1080/15732479.2018.1496119
25. Saydam D, Bocchini P, Frangopol DM (2013) Time-dependent risk associated with deterioration of highway bridge networks. Eng Struct 54:221–233. https://doi.org/10.1016/j.engstruct.2013.04.009
26. Yang DY, Frangopol DM (2018) Risk-informed bridge ranking at project and network levels. J Infrastruct Syst 24:04018018. https://doi.org/10.1061/(ASCE)IS.1943-555X.0000430
27. Bocchini P, Frangopol DM (2011) A stochastic computational framework for the joint transportation network fragility analysis and traffic flow distribution under extreme events. Probabilist Eng Mech 26:182–193. https://doi.org/10.1016/j.probengmech.2010.11.007
28. Capacci L, Biondini F (2020) Probabilistic life-cycle seismic resilience assessment of aging bridge networks considering infrastructure upgrading. Struct Infrastruct E 16:659–675. https://doi.org/10.1080/15732479.2020.1716258
29. Akiyama M, Frangopol DM, Arai M, Koshimura S (2013) Reliability of bridges under tsunami hazards: Emphasis on the 2011 Tohoku-Oki earthquake. Earthq Spectra 29:295–314. https://doi.org/10.1193/1.4000112
30. Ataei N, Padgett JE (2013) Probabilistic modeling of bridge deck unseating during hurricane events. J Bridge Eng 18:275–286. https://doi.org/10.1061/(ASCE)BE.1943-5592.0000371
31. Kameshwar S, Padgett JE (2014) Multi-hazard risk assessment of highway bridges subjected to earthquake and hurricane hazards. Eng Struct 78:154–166. https://doi.org/10.1016/j.engstruct.2014.05.016
32. Chang CM, Ortega O, Weidner J (2021) Integrating the risk of climate change into transportation asset management to support bridge network-level decision-making. J Infrastruct Syst 27:04020044. https://doi.org/10.1061/(ASCE)IS.1943-555X.0000590
33. Jayaram N, Baker JW (2009) Correlation model for spatially distributed ground-motion intensities. Earthq Eng Struct D 38:1687–1708. https://doi.org/10.1002/eqe.922
34. Liu L, Frangopol DM, Mondoro A, Yang DY (2018) Sustainability-informed bridge ranking under scour based on transportation network performance and multiattribute utility. J Bridge Eng 23:04018082. https://doi.org/10.1061/(ASCE)BE.1943-5592.0001296
35. Zhang W, Wang N (2016) Resilience-based risk mitigation for road networks. Struct Saf 62:57–65. https://doi.org/10.1016/j.strusafe.2016.06.003
36. Christou V, Bocchini P, Miranda MJ, Karamlou A (2018) Effective sampling of spatially correlated intensity maps using hazard quantization: Application to seismic events. ASCE-ASME J Risk Uncertainty Eng Syst, Part A: Civ Eng 4:04017035. https://doi.org/10.1061/AJRUA6.0000939
37. Abarca A, Monteiro R, O'Reilly GJ (2022) Simplified methodology for indirect loss–based prioritization in roadway bridge network risk assessment. Int J Disast Risk Re 74:102948. https://doi.org/10.1016/j.ijdrr.2022.102948
38. Liu T, Meidani H (2024) End-to-end heterogeneous graph neural networks for traffic assignment. Transp Res Part C Emerg Technol 165:104695. https://doi.org/10.1016/j.trc.2024.104695
39. Li P, Wang Z, Zhao B, Becker T, Soga K (2025) Surrogate modeling for identifying critical bridges in traffic networks under earthquake conditions. Transport Res D-Tr E 138:104512. https://doi.org/10.1016/j.trd.2024.104512
40. Yang DY (2022) Adaptive risk-based life-cycle management for large-scale structures using deep reinforcement learning and surrogate modeling. J Eng Mech 148:04021126. https://doi.org/10.1061/(ASCE)EM.1943-7889.0002028
41. Nabian MA, Meidani H (2018) Deep learning for accelerated seismic reliability analysis of transportation networks. Comput-Aided Civ Inf 33:443–458. https://doi.org/10.1111/mice.12359
42. Liu T, Meidani H (2023) Optimizing seismic retrofit of bridges: Integrating efficient graph neural network surrogates and transportation equity. In: Proceedings of Cyber-Physical Systems and Internet of Things Week 2023. Association for Computing Machinery, New York, NY, USA, pp 367–372
43. Liu T, Meidani H (2024) Graph neural network surrogate for seismic reliability analysis of highway bridge systems. J Infrastruct Syst 30:05024004. https://doi.org/10.1061/JITSE4.ISENG-2264
44. Shim CS, Dang NS, Lon S, Jeon CH (2019) Development of a bridge maintenance system for prestressed concrete bridges using 3D digital twin model. Struct Infrastruct E 15:1319–1332. https://doi.org/10.1080/15732479.2019.1620789
45. Franciosi M, Kasser M, Viviani M (2024) Digital twins in bridge engineering for streamlined maintenance and enhanced sustainability. Automat Constr 168:105834. https://doi.org/10.1016/j.autcon.2024.105834
46. Adibfar A, Costin AM (2022) Creation of a mock-up bridge digital twin by fusing intelligent transportation systems (ITS) data into bridge information model (Br IM). J Constr Eng M ASCE 148:04022094. https://doi.org/10.1061/(ASCE)CO.1943-7862.0002332
47. Chacón R, Ramonell C, Posada H, Sierra P, Tomar R, Martínez De La Rosa C, Rodriguez A, Koulalis I, Ioannidis K, Wagmeister S (2024) Digital twinning during load tests of railway bridges - case study: the high-speed railway network, Extremadura, Spain. Struct Infrastruct E 20:1105–1119. https://doi.org/10.1080/15732479.2023.2264840
48. Guo X, Fang S (2024) Digital twin based lifecycle modeling and state evaluation of cable-stayed bridges. Eng Comput-Germany 40:885–899. https://doi.org/10.1007/s00366-023-01835-6
49. Costin A, Adibfar A, Bridge J (2024) Digital twin framework for bridge structural health monitoring utilizing existing technologies: New paradigm for enhanced management, operation, and maintenance. Transport Res Rec 2678:1095–1106. https://doi.org/10.1177/03611981231208908
50. Braik AM, Koliou M (2023) A novel digital twin framework of electric power infrastructure systems subjected to hurricanes. Int J Disast Risk Re 97:104020. https://doi.org/10.1016/j.ijdrr.2023.104020
51. Braik AM, Koliou M (2024) A digital twin framework for efficient electric power restoration and resilient recovery in the aftermath of hurricanes considering the interdependencies with road network and essential facilities. Resil Cities Struct 3:79–91. https://doi.org/10.1016/j.rcns.2024.07.004
52. Torzoni M, Tezzele M, Mariani S, Manzoni A, Willcox KE (2024) A digital twin framework for civil engineering structures. Comput Methods Appl Mech Eng 418:116584. https://doi.org/10.1016/j.cma.2023.116584
53. Decò A, Frangopol DM (2013) Life-cycle risk assessment of spatially distributed aging bridges under seismic and traffic hazards. Earthq Spectra 29:127–153. https://doi.org/10.1193/1.4000094
54. Dong Y, Frangopol DM (2015) Risk and resilience assessment of bridges under mainshock and aftershocks incorporating uncertainties. Eng Struct 83:198–208. https://doi.org/10.1016/j.engstruct.2014.10.050
55. Cheng M, Frangopol DM, Gong C (2020) Acceptance of corrosion-resistant steel in design of steel girder bridges based on expected utility theory. J Bridge Eng 25:04020098. https://doi.org/10.1061/(ASCE)BE.1943-5592.0001627
56. Stein SM, Young GK, Trent RE, Pearson DR (1999) Prioritizing scour vulnerable bridges using risk. J Infrastruct Syst 5:95–101. https://doi.org/10.1061/(ASCE)1076-0342(1999)5:3(95)
57. Dong Y, Frangopol DM, Saydam D (2013) Time-variant sustainability assessment of seismically vulnerable bridges subjected to multiple hazards. Earthq Engng Struct Dyn 42:1451–1467. https://doi.org/10.1002/eqe.2281
58. Shamo B, Asa E, Membah J (2015) Linear spatial interpolation and analysis of annual average daily traffic data. J Comput Civ Eng 29:04014022. https://doi.org/10.1061/(ASCE)CP.1943-5487.0000281
59. American Association of State Highway and Transportation Officials (1977) A manual on user benefit analysis of highway and bus-transit improvements. AASHTO, Washington, D.C.
60. Ghosn M, Moses F, Frangopol DM (2009) Redundancy and robustness of highway bridge superstructures and substructures. Struct Infrastruct E 6:257-278
61. Pourret O, Na P, Marcot B (2008) Bayesian networks: a practical guide to applications. John Wiley & Sons
62. Schulte O, Frigo G, Greiner R, Wei L, Khosravi H (2009) A new hybrid method for Bayesian network learning with dependency constraints. In:

2009 IEEE Symposium on Computational Intelligence and Data Mining. IEEE, New York, NY, USA, pp 53–60
63. Scutari M, Graafland CE, Gutiérrez JM (2019) Who learns better Bayesian network structures: Accuracy and speed of structure learning algorithms. Int J Approx Reason 115:235–253. https://doi.org/10.1016/j.ijar.2019.10.003
64. Tsamardinos I, Brown LE, Aliferis CF (2006) The max-min hill-climbing Bayesian network structure learning algorithm. Mach Learn 65:31–78. https://doi.org/10.1007/s10994-006-6889-7
65. Laitila P, Virtanen K (2016) Improving construction of conditional probability tables for ranked nodes in Bayesian networks. IEEE Trans Knowl Data Eng 28:1691–1705. https://doi.org/10.1109/TKDE.2016.2535229
66. Ramoni M, Sebastiani P (1997) Learning Bayesian networks from incomplete databases. In: Proceedings of the 13th Conference on Uncertainty in Artificial Intelligence. Morgan Kaufmann, San Francisco, USA, pp 401—408
67. Jensen FV, Nielsen TD (2007) Bayesian networks and decision graphs. Springer, New York
68. Bensi M, Der Kiureghian A, Straub D (2011) Bayesian network modeling of correlated random variables drawn from a Gaussian random field. Struct Saf 33:317–332. https://doi.org/10.1016/j.strusafe.2011.05.001
69. Cheng M, Shah SM, Nanni A, Gao HO (2024) Automated knowledge graphs for complex systems (AutoGraCS): Applications to management of bridge networks. Resil Cities Struct 4:95–106
70. Beuzen T, Marshall L, Splinter KD (2018) A comparison of methods for discretizing continuous variables in Bayesian Networks. Environ Modell Softw 108:61–66. https://doi.org/10.1016/j.envsoft.2018.07.007
71. Cheng M, Gao HO (2023) Data-driven life-cycle risk assessment of bridge networks using Bayesian network. In: Life-cycle of Structures and Infrastructure Systems, 1st edn. CRC Press, London, pp 1953–1960
72. FHWA (Federal Highway Administration) (1992) National Bridge Inventory (NBI). US Department of Transportation. https://www.fhwa.dot.gov/bridge/nbi.cfm. Accessed 10 Feb 2025
73. FDOT (Florida Department of Transportation) (2023) Annual Average Daily Traffic (AADT). Florida Department of Transportation Open Data Hub. https://gis-fdot.opendata.arcgis.com/. Accessed 10 Feb 2025
74. USGS (United States Geological Survey) (2025) USGS Water Resources. United States Geological Survey National Water Information System. https://waterdata.usgs.gov/nwis/rt. Accessed 10 Feb 2025
75. Cheng M, Frangopol DM (2023) Efficient scenario analysis for optimal adaptation of bridge networks under deep uncertainties through knowledge transfer. Struct Saf 100:102278. https://doi.org/10.1016/j.strusafe.2022.102278
76. FHWA (Federal Highway Administration) (2022) National Bridge Inspection Standards (NBIS). US Department of Transportation. https://www.fhwa.dot.gov/bridge/nbis.cfm. Accessed 10 Feb 2025
77. Obeysekera J, Irizarry M, Park J, Barnes J, Dessalegne T (2011) Climate change and its implications for water resources management in south Florida. Stoch Environ Res Risk Assess 25:495–516. https://doi.org/10.1007/s00477-010-0418-8
78. Bocchini P, Frangopol DM (2012) Optimal resilience- and cost-based post-disaster intervention prioritization for bridges along a highway segment. J Bridge Eng 17:117–129. https://doi.org/10.1061/(ASCE)BE.1943-5592.0000201
79. Yang DY, Frangopol DM (2019) Life-cycle management of deteriorating civil infrastructure considering resilience to lifetime hazards: A general approach based on renewal-reward processes. Reliab Eng Syst Safe 183:197–212. https://doi.org/10.1016/j.ress.2018.11.016
80. Sharma N, Tabandeh A, Gardoni P (2018) Resilience analysis: a mathematical formulation to model resilience of engineering systems. Sustain Resil Infras 3:49–67. https://doi.org/10.1080/23789689.2017.1345257

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.